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Key Technologies for Machine Vision for Picking Robots: Review and Benchmarking

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Abstract: The increase in precision agriculture has promoted the development of picking robot technology, and the visual recognition system at its core is crucial for improving the level of agricultural automation. This paper reviews the progress of visual recognition technology for picking robots, including image capture technology, target detection algorithms, spatial positioning strategies and scene understanding. This article begins with a description of the basic structure and function of the vision system of the picking robot and emphasizes the importance of achieving high-efficiency and high-accuracy recognition in the natural agricultural environment. Subsequently, various image processing techniques and vision algorithms, including color image analysis, three-dimensional depth perception, and automatic object recognition technology that integrates machine learning and deep learning algorithms, were analysed. At the same time, the paper also highlights the challenges of existing technologies in dynamic lighting, occlusion problems, fruit maturity diversity, and real-time processing capabilities. This paper further discusses multisensor information fusion technology and discusses methods for combining visual recognition with a robot control system to improve the accuracy and working rate of picking. At the same time, this paper also introduces innovative research, such as the application of convolutional neural networks (CNNs) for accurate fruit detection and the development of event-based vision systems to improve the response speed of the system. At the end of this paper, the future development of visual recognition technology for picking robots is predicted, and new research trends are proposed, including the refinement of algorithms, hardware innovation, and the adaptability of technology to different agricultural conditions. The purpose of this paper is to provide a comprehensive analysis of visual recognition technology for researchers and practitioners in the field of agricultural robotics, including current achievements, existing challenges and future development prospects.

Keywords: Picking robot, visual system, perception technology, image processing, machine learning, deep learning.

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1 Introduction

In the rapid development of modern agriculture, picking robots, as a key technology to improve production efficiency, reduce labor costs, and improve operation quality are gradually becoming the focus of research[1, 2]. By simulating human perception, decision making and action processes, these intelligent systems are able to identify and pick ripe fruits from a variety of crops[3, 4] while avoiding damage to crops or missed picking. As the core component of the picking robot, the vision system undertakes the important task of perceiving and understanding the environment[5, 6]. In addition to accurately identifying target fruits in changing natural environments, large amounts of data must be processed in real time to enable efficient and precise farming operations[7–9].

However, due to the complexity of the agricultural environment, the vision system of picking robots faces many challenges[10, 11]. The changes in lighting conditions, the shading between fruits, and the differences in the appearance of fruits with different maturities all necessitate greater requirements for the recognition ability of the visual system[12]. In addition, real-time requirements also challenge the processing speed and algorithm efficiency of the system. To address these challenges[13], researchers are constantly exploring new image processing techniques and computer vision algorithms[14], such as using machine learning and deep learning to improve recognition accuracy, using multisensor data fusion to enhance environmental awareness, and developing algorithms with greater real-time performance to meet the needs of dynamic picking.

In this paper, we review the composition and function of the vision system of picking robots, discuss the current progress of image processing and object recognition technology, analyse the challenges of existing technology, and look forward to the future direction of technology development. Through this review, we aim to provide researchers and engineers in the field with a comprehensive

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analysis of the current state of research and future trends to promote the further development and application of picking robot technology.

2 Target detection and recognition

Recognizing the target fruit is not only necessary for the picking robot but also a crucial factor that affects its performance[15]. When operating in an unstructured environment, target fruit recognition is influenced by various interference conditions[16]. Therefore, automatic target fruit recognition has become one of the key problems in visual control. To address this issue, both domestic and international scholars have proposed numerous recognition methods, some of which have been applied to picking robots[17, 18]. Based on a summary of these methods, they can be categorized into the following three types: single feature analysis method, multifeature fusion analysis method, and machine learning-based method[19–21]. As shown in Fig. 1, we show the picking process of the picking robot.

2.1 Traditional digital image processing technology

Color[22–24], shape[25, 26], and texture[27] are important features used by picking robots to distinguish fruits from background images[28]. Researchers have utilized feature segmentation of base color[29], shape[30], and texture[31] to detect fruits[32], such as apples[33], tomatoes[34], oranges[35], and so on. Fruits exhibit significant and stable color features and have little dependence on image size[36]. However, when using color features to detect fruit, factors such as fruit maturity[37], color variation[38], fruit variety[39], background features[40], and changes in external light can influence the detection accuracy. Geometry provides another set of characteristic indicators for fruit detection. For example, apples and citrus fruits are typically rounder than branches and leaves. Mature green apples may blend in with the background image, making them easier to detect using geometric features. Generally, geometric features are less affected by lighting conditions, making them suitable for feature extraction in orchard environments. However, the main challenge of using shape feature segmentation to detect fruit is the unstructured nature of orchard environments[41]. Fruits may be obscured by branches and leaves or clustered together, resulting in changes in geometric parameters such as fruit shape and size, which can reduce recognition accuracy[42]. Texture is another feature used to distinguish fruit from other ground images. The fruit surface is often smoother than the background surface (branches, leaves, trunk). Texture-based detection can first separate surfaces with the same texture and then identify the target fruit based on the edges of the separated surface[43]. Because fruit surface texture is not affected by color, texture features are often used to detect fruits with colors similar to those of branches and leaves. However, changes in texture due to varying illumination, complex backgrounds, and different fruit maturity levels can limit the accuracy of segmentation and detection based on traditional features. To address the limitations of single feature segmentation detection, integrating two or more features to form a multifeature fusion detection method can effectively improve the detection accuracy and robustness[44–46]. The comparison of experimental results of digital image processing technology is shown in Table 1.

1) Color feature-based fruit recognition. Color features are extensively utilized for distinguishing mature fruits from complex backgrounds in the development of picking robots[47]. The core principle of target fruit recognition based on color features involves segmenting all pixels in the image into two groups using a pixel threshold: One group corresponds to the target fruit image[48–50], while the other represents the background im-

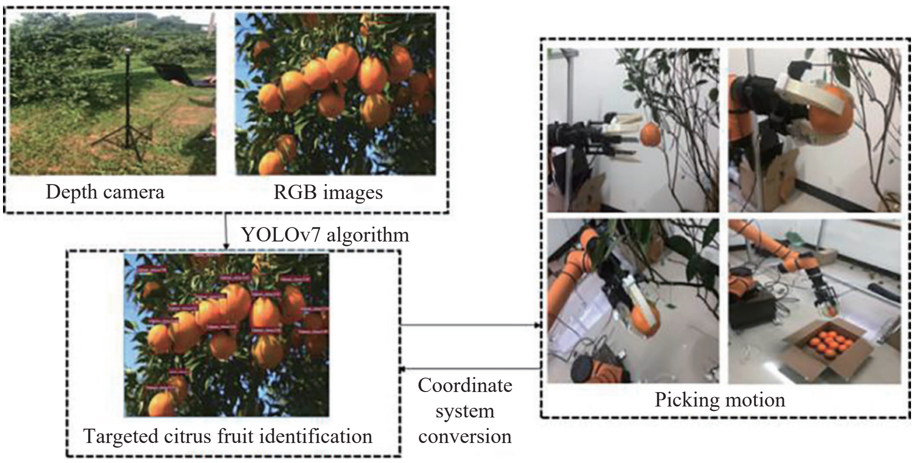

Fig. 1 Framework of an autonomous picking system for picking robots (Colored figures are available in the online version at https://link.springer.com/journal/11633)

Table 1 Comparison of experimental results of digital image processing technology

Traditional digital image technology	Advantage	Accuracy (%)	Limitations	Applied crops
Color features	It can significantly distinguish between fruit object and background	80 – 85	Affected by light conditions	Apple, citrus
Geometric features	It can obtain the contour information of fruit object	80 – 87	Affected by occlusion and fruit size	Tomato, green citrus, green apple
Texture feature	It can significantly separate fruit and background information	75 – 90	It is affected by light conditions, occlusion and fruit clustering	Pineapple, balsam pear
Multifeature fusion	Make up for the limitation of single feature and improve the recognition reliability in complex natural environment	89 – 92	It is affected by light conditions, occlusion and fruit clustering	Oranges, apples, mangoes

age[51, 52]. However, selecting the segmentation threshold is critical and can be influenced by factors such as changes in lighting conditions[53], which may impact recognition success rates[54, 55]. To mitigate the effects of light changes, researchers have proposed various color models, including HIS, Lab*, and LCD, to extract color features[56]. For example, Arefi et al.[57] proposed a tomato recognition algorithm that combines RGB, HIS and YIQ color space models, achieving a successful recognition rate of 96.36% under greenhouse conditions with artificial lighting[58]. Similarly, Liu et al.[59] introduced a DRG DRB color coding model for apple segmentation from the background, achieving a success rate of 90%[60]. Furthermore, Wei et al.[61] proposed a fruit recognition method utilizing color features that integrates the Otsu threshold segmentation algorithm and Ohta color model features[62]. The Ohta color model, derived through a linear transformation of the RGB color model, yielded significantly higher recognition rates than the RGB color model, reaching up to 95%[63]. However, when the fruit color closely resembles the background color, recognition methods that rely solely on color features often exhibit low recognition rates, limiting their practical application[64].

2) Geometric features. Geometric features play a crucial role in identifying target fruits, as they are independent of color information, thereby effectively mitigating the influence of light and color interference[63]. For instance, Oualid Djekoune et al.[65] introduced a detection algorithm based on an improved Hough circle transformation in the 1980s, leveraging roundness feature information to effectively identify normal tomatoes obscured by branches and leaves. Similarly, Xie et al.[66] explored a recognition algorithm based on concave search and the Hough transform, which enhanced the recognition rate of strawberries, particularly when the occlusion rate was less than $\frac{1}{2}$. Moreover, Tene-Cohen et al.[67] devised an automatic apple recognition algorithm for trees utilizing convex geometric features. Initially, the edge information of apples was extracted, followed by the identification of convex apples using the Canny edge detection operator, resulting in the successful recognition of 94% of the apples on visible trees[68].

3) Texture features. Texture, an essential physical property of fruits and vegetables, serves as a valuable feature for distinguishing target fruits from background areas in images, particularly in the presence of interference such as adhesion or occlusion[69]. For instance, Kurtulmus et al.[70] introduced an eigenfruit recognition algorithm based on Gabor texture features and successfully applied it to the automatic recognition of tomatoes. However, it is worth noting that the extraction of texture features often entails substantial computational requirements, necessitating the optimization of computer hardware and algorithms. Furthermore, textural features are frequently integrated with color and geometric features to enhance recognition accuracy[71].

4) Multifeature fusion. While single-feature analysis methods can detect fruit in natural environments, variations in lighting conditions and obstructive elements such as branches, leaves, and clustered growth can distort the geometric features of fruit in images. Consequently, relying solely on a single feature for detection may not yield optimal results[72]. Research has shown that integrating two or more features into multiple features can significantly enhance the detection accuracy and robustness. Currently, most image processing techniques used by researchers for fruit detection and recognition require setting thresholds based on color, shape, or texture features. However, determining the optimal threshold poses a challenge because it often varies with age. To address this issue, Payne and Glen[73] proposed a method utilizing RGB and YCbCr color segmentation and texture segmentation[74]. By leveraging the variability of adjacent pixels, this approach classifies pixels into target fruit and background categories, facilitating fruit detection and recognition. However, this algorithm heavily relies on image color features, leading to reduced recognition accuracy when color features are not prominent[75]. To mitigate this limitation, Payne and Glen[73] introduced boundary-constrained mean and edge detection filters based on a previously proposed algorithm. This modification reduces the dependence on color features while enhancing the utilization of texture filtering, resulting in significantly improved detection performance com-

pared to that of the original algorithm[76]. Multifeature fusion and machine learning techniques provide avenues for enhancing the accuracy of single-fruit detection and streamline the cumbersome process of adjusting thresholds in fruit detection algorithms. By integrating multiple features and leveraging machine learning, researchers can enhance detection accuracy while minimizing the need for manual threshold adjustments, thereby advancing the capabilities of fruit detection and recognition systems[77].

2.2 Machine learning methods

Machine learning represents a significant advancement in artificial intelligence research, leading to rapid developments in robot learning since the 1980s. Various machine learning techniques have been applied to target fruit recognition in picking robots. The main types of multimodal image recognition algorithms are shown in Table 2.

1) Clustering algorithm based on K-means. The K-means algorithm, first proposed in 1955, primarily determines the value of K in advance: It is the simplest and most widely used clustering algorithm, and randomly selected centers for K-means clustering can represent various disciplines effectively. Despite its simplicity, it significantly influences the final classification results[82]. Researchers continue to explore and refine this algorithm due to its ease of implementation, high efficiency, and effectiveness. In fruit recognition, K-means clustering is extensively utilized. Building upon the K-means algorithm, Yamamoto[83] introduced an X-means clustering algorithm to address the challenge of clustered fruit recognition. Xiao et al.[84] proposed a fuzzy clustering method specifically tailored for identifying clustered tomatoes, demonstrating superior performance compared to traditional K-means clustering and Otsu threshold segmentation algorithms. Luo et al.[85] employed the K-means clustering method for segmenting, denoising, and filling captured images to obtain complete and closed target image regions[86]. Wang et al.[87] further enhanced the algorithm′s robustness against light interference by improving wavelet transforms and integrating the K-means clustering method for image segmentation. This approach not only effectively mitigates the impact of light changes but also accurately segments fruits of different colors. Additionally, Moallem et al.[88] leveraged the K-means clustering method to detect the Calyx region of Cb components in the YCbCr color space. They employed a multilayer perceptron neural network for defect segmentation, extracting statistical, textural, and geometric features from the defect segmentation region. Despite its high accuracy in recognition and classification, the K-means algorithm has its limitations[89].

2) Bayesian classifier. The Bayesian classifier is a widely adopted supervised classification method that minimizes the Bayesian risk and probability error or maximizes the posteriori probability. For instance, Harrell et al.[90] utilized a Bayesian classifier to identify citrus fruits in natural settings. The classifier model incorporated chromaticity and intensity information from the images, achieving a pixel-level recognition rate of 75%. Min et al.[91] employed a naive Bayesian classifier to distinguish between fruit and nonfruit regressions. Although this algorithm effectively mitigated color similarity interference between green tomatoes and background foliage, it required prior probability information from training images. However, variations in fruit surface pixel values under different lighting conditions impacted the accuracy, resulting in an 86% accuracy rate. Notably, the algorithm demonstrated rapid recognition and classification speeds, performed well on small-scale data, combined multiple classification tasks, and supported incremental training. However, this approach necessitated the calculation of prior probabilities and exhibited a certain error rate in classification decisions. Moreover, the al-

Table 2 The main kinds of multimodal image recognition algorithms

Machine learning method	Advantage	Recognition accuracy (%)	Limitations	Applied crops
Clustering algorithm based on K-means	Short computing time, fast response speed, good clustering effect, automatic division of target fruit and background image	80 – 90	It is sensitive to heterogeneous data and needs to set K-value in advance. The solution of K-value is complex	Litchi[78], grape
Bayesian classifier	Small datasets perform well, the classification process is simple, the response speed is fast, and they are equally effective in multi classification problems	75 – 80	For the variable features that have not appeared in the training set, the recognition function fails, and the priori probability is affected by the illumination	Tomato[79]
KNN based clustering algorithm	High precision and insensitive to outliers	85 – 90	Appropriate K-value needs to be set, and the size of K-value has a great impact on the recognition accuracy	Mango[80]
SVM	It can classify the data outside the training set image well, with simple calculation and high recognition accuracy	90 – 93	For the new dataset, the algorithm parameters and kernel function parameters need to be readjusted	Apple, banana[81]

gorithm might fail to recognize data not present in the training set.

3) KNN-based clustering algorithm. The K-nearest neighbors (KNN) clustering algorithm is a supervised learning method frequently applied in classification and regression models. It classifies unknown feature vectors based on the attributes of the K-nearest neighbors from existing training samples. Researchers often employ the KNN clustering algorithm to classify various fruits. For instance, Seng utilized KNN to classify and recognize fruit images based on color, shape, and size features. The method demonstrated high recognition and classification accuracy, insensitivity to outliers, and no assumptions about the data. However, it incurred a high computational burden, exhibited high time and space complexity, and was susceptible to environmental factors such as lighting conditions and fruit size.

4) SVM-based algorithms. A support vector machine (SVM) is a feedforward neural network that is a supervised statistical learning algorithm commonly used in linear and nonlinear regression analysis and pattern classification. SVM classifiers excel in fruit image segmentation under strong illumination, outperforming other classifiers. Gao et al.[92] employed morphological operations and multiclass SVM to segment fruits and branches simultaneously, achieving a recognition accuracy of 92% for citrus fruits. Xiao et al.[84] introduced an SVM classifier after extracting color and shape features from images, reducing the recognition time to 352 milliseconds. However, this method can only recognize undamaged target fruits. To address occlusion, irregular shapes, and background similarity, Gill et al.[93] proposed a method combining SVM texture classification, Canny edge detection, a graph-based connected component algorithm, and Hough line detection for fruit image detection. Furthermore, Peng et al.[94] utilized an SVM classifier to classify and recognize six types of fruits, achieving a recognition accuracy exceeding 90%. SVM classifiers address the challenge of small sample sizes in natural environments and maintain computational efficiency when mapping to high-dimensional spaces. However, these methods exhibit high accuracy primarily in binary classification tasks and are sensitive to missing datasets and kernel functions[95].

2.3 Convolutional neural network with deep learning

Convolutional neural networks (CNNs) represent multilayer feedforward neural network models with convolution operations, allowing for the extraction of useful features from locally related data points using a set of convolution kernels in each layer. CNNs learn through the backpropagation algorithm, optimizing the objective function based on a brain-like learning mechanism inspired by human responses. The continuous success of the backpropagation algorithm and CNNs has propelled artificial intelligence into a new stage of development. Deep architectures often outperform shallow architectures when tackling complex learning problems. Following the success of the LeNet convolutional neural network model on the MNIST dataset, various relevant network models, such as AlexNet, VGG, ResNet, MobileNet, Fast R-CNN, SSD and YOLO, have emerged, with wide applications in crop image processing, object segmentation, and other domains[96].

1) VGGNet network model. The VGG convolutional neural network series was proposed by the Oxford University research team during the Large Scale Visual Recognition Challenge in 2014. "VGG" stands for the visual geometry group at Oxford University. The VGG series includes models such as VGG-11, VGG-13, VGG-16 and VGG-19. The main improvements in VGG areas are as follows:

i) A VGG can be constructed with up to 19 layers, significantly increasing the network depth.

ii) It reduces the size of the convolution kernels and employs only 3×3 kernels with a stride of 1 and a padding of 1. Compared to AlexNet, which uses 11×11, 5×5, and 3×3 kernels, VGG's use of smaller kernels leads to fewer parameters and lower computational costs.

iii) VGG employs several small convolution kernels (3×3) instead of larger kernels (5×5 or 11×11), demonstrating better performance. This approach reduces the number of parameters, resulting in lower computational complexity while improving performance by deepening the network structure. While VGG achieves impressive results in age classification, it utilizes more parameters and requires more computing resources, posing challenges for deployment on low-resource systems. Hence, there is an urgent need to explore new algorithms and develop parallel computing systems to address these limitations. The VGG network model is shown in Fig. 2.

2) SSD network model. The single shot multibox detector (SSD) network model represents a single-stage detection model that integrates the VGGNet and ResNet architectures. It excels in both accuracy and speed for target detection and recognition tasks. SSD operates by performing detection directly on feature maps at multiple scales, enabling it to efficiently detect objects of various sizes. This approach eliminates the need for multiple region proposals and subsequent refinement stages, making SSD a single-stage detector. By leveraging both the VGGNet and ResNet architectures, SSD benefits from the strengths of these models, including robust feature extraction capabilities and improved performance in object detection tasks. This function of architectures enhances the overall accuracy and efficiency of the SSD model, making it a powerful tool for real-time object detection applications. The SSD network model is shown in Fig. 3.

Peng et al.[97] introduced an enhanced detection algorithm based on the single shot multibox detector (SSD)

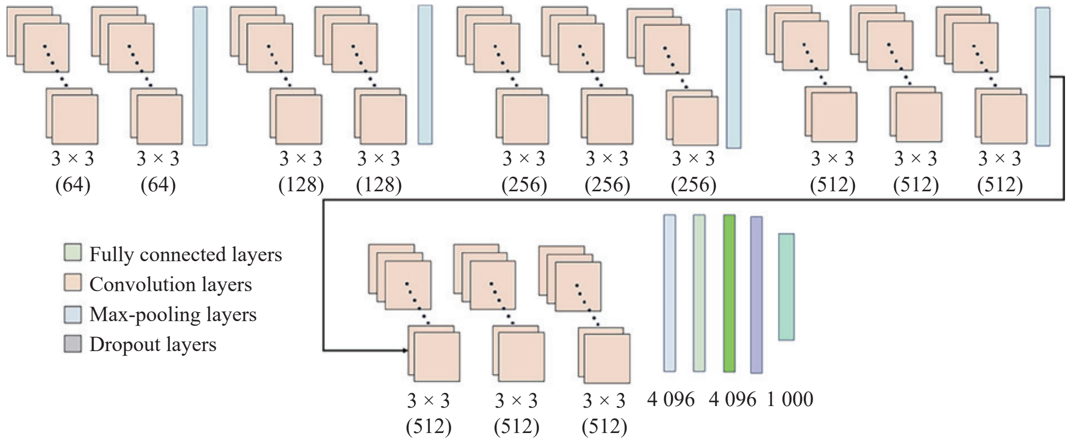

Fig. 2 Diagram of the VGG network model (Colored figures are available in the online version at https://link.springer.com/journal/11633)

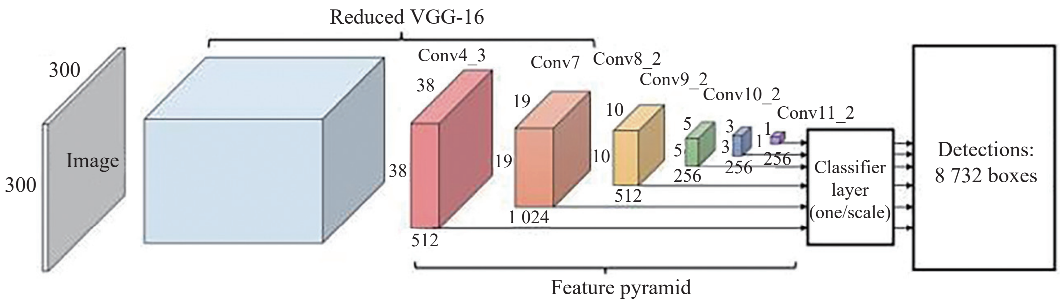

Fig. 3 Diagram of the SSD network model

network to achieve real-time mango detection. Building upon the SSD network architecture, the proposed algorithm outperforms the VGG-16 and ZFNet-based SSD models. Despite exhibiting superior accuracy compared to faster CNNs, the enhanced algorithm faces challenges due to its deep layers and large parameter count, resulting in slow operation speeds. To address these concerns, Xiao et al.[98] proposed the SSD MobileNet algorithm, which replaces the VGG-16 network with the decomposable convolutional MobileNet network. This modification significantly enhances the recognition speed of the model while maintaining high accuracy in fruit detection tasks. Building upon this advancement, Peng et al.[97] proposed further improvements by replacing the VGG-16 input model in the traditional SSD deep learning model with the ResNet101 model. The resulting enhanced SSD deep learning fruit detection model demonstrates superior detection accuracy across various environments. Moreover, the improved model exhibits excellent generalizability and robustness, enabling accurate detection of numerous types of fruits in natural settings[99].

3) YOLO network model. YOLOv3, the latest iteration in the you only look once (YOLO) series, builds upon the foundational concepts and methodologies of its predecessors, YOLOv1 and YOLOv2, to introduce significant enhancements in object detection capabilities. One of the key advancements in YOLOv3 is the utilization of the location box prediction method from YOLOv2, which enhances the accuracy of object localization. Additionally, YOLOv3 introduces the Darknet-53 network architecture, providing a deeper and more powerful backbone for feature extraction. Another noteworthy improvement is the replacement of the softmax classification method with the logistic method. This change enhances the model's ability to classify objects by assigning probabilities to each class independently, resulting in more accurate predictions. Furthermore, YOLOv3 incorporates a multiscale prediction approach, allowing it to effectively detect small and micro targets while maintaining a high detection speed. This feature is particularly advantageous in scenarios where objects of varying sizes need to be detected simultaneously. In summary, YOLOv3 represents a significant advancement in object detection technology, offering improved prediction capabilities, enhanced classification accuracy, and efficient detection of objects across different scales, thereby expanding its application various domains[100].

3 Multisensor information fusion

In the field of picking robots, vision sensors play a crucial role in capturing rich image information about files and fruits. However, because these sensors usually do not contain depth information, they are limited in obtaining accurate three-dimensional spatial data and are susceptible to interference from environmental factors such as changes in lighting. LiDAR, meanwhile, is known for its excellent robustness and accurate three-dimensional space measurement capabilities despite its shortcomings in identifying specific categories of targets. In the actual operation of agricultural robots, due to the variability of the operating environment, it is often difficult for a single sensor to provide consistently reliable detection results in all cases. To overcome this challenge, researchers are beginning to explore multisensor fusion techniques to enhance robots′ environmental awareness. The multisensor obstacle detection system, as shown in Fig. 4, consists of four key units: data acquisition, data preprocessing, information fusion and decision making. According to the different stages of fusion, the information fusion unit can be divided into three levels: the original data level, the feature level and the decision level. Raw data-level fusion involves directly combining raw data collected by different sensors, which is usually achieved by direct fusion or weighted fusion methods. The feature-level function is further processed by first extracting features from each sensor dataset and then performing feature matching, association and clustering, which helps to reduce the complexity of the data and improve the processing efficiency. Decision-level fusion, on the other hand, synthesizes the decision results after each sensor completes the decision independently. Although this method has high robustness and a rapid response, it may also be affected by the error reported for a single sensor. The original data-level and feature-level fusion are early fusion methods and depend on prior knowledge. Decision-level fusion is a type of late fusion based on already formed decisions. Ultimately, the diagnostic ultrasound (DUS) filters out reliable results based on the fused information and passes critical information about obstacles – including their location, category, size, speed and heading – to other parts of the automated system, such as path planning or control modules, for a more precise and automated workflow. Through this multilevel, multistage integration strategy, the picking robot can work more intelligently and adaptively, improving the efficiency and safety of the operation[101]. Multi-sensor information fusion processing is shown in Fig. 5.

Multisensor fusion technology plays a key role in improving the robustness and accuracy of sensing systems,

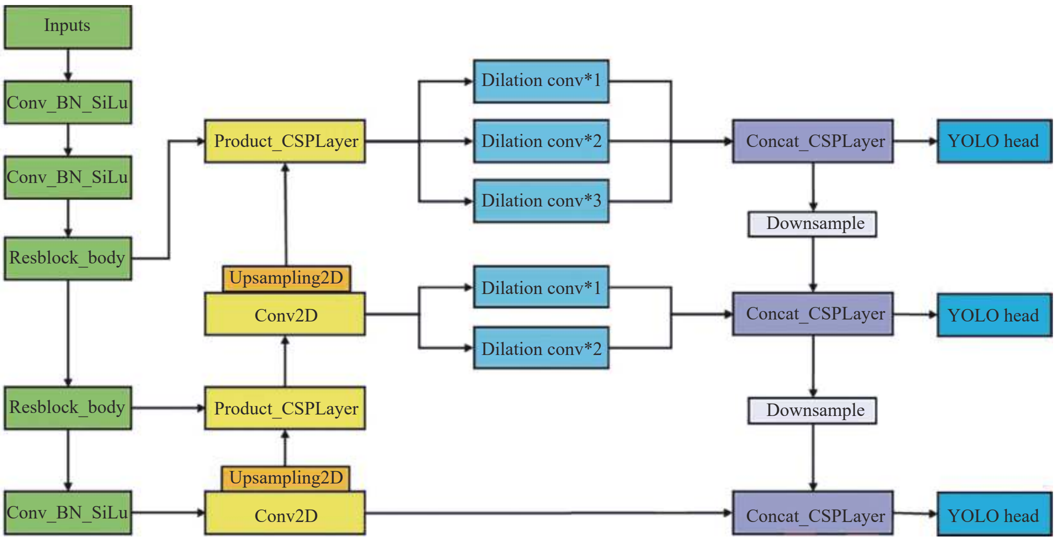

Fig. 4 Diagram of the YOLOv4 network model

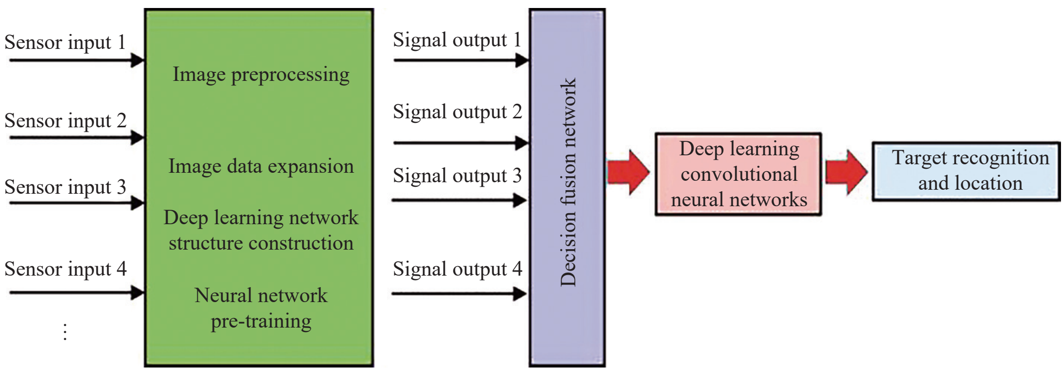

Fig. 5 Multisensor information fusion process

which has been demonstrated in numerous studies. By cleverly combining redundant information and complementary information, this technology achieves depth enhancement of environmental perception. The core of redundant fusion is to update the confidence degree of the fused information through data identification and association. This process not only improves the accuracy of the measurement results but also significantly reduces the uncertainty of the data. It relies on multiple independent measurements of the same phenomenon, which confirm each other and collectively contribute to a more accurate comprehensive view. The integration of complementary information further broadens the vision of the perceptual system. It is divided into two levels: The first is the application of different types of physical measurements within a unified field of view, such as distance, color and temperature, and each measurement provides a unique perspective of the environment; The second is the construction of a more comprehensive and detailed environmental model by integrating information from different fields of view. In the specific application of fruit and vegetable detection, the multisensor fusion process is particularly complex. We must not only select the appropriate sensor combination but also carefully adjust and optimize the fusion strategy for specific application scenarios. This approach involves a deep understanding and intelligent processing of sensor data to ensure that fruit can be accurately detected and identified in a variable agricultural environment. In summary, multisensor fusion technology has great potential for improving the accuracy and reliability of fruit detection. As technology continues to evolve and innovate, we believe that this technique will play an increasingly important role in the field of smart agriculture. Multi-sensor fusion is shown in Fig. 6.

4 Challenges and prospects

With the rapid advancement in the fields of computer vision and machine learning, particularly the exceptional performance of deep learning across various applications, there has been a significant enhancement in the environmental and mental perception capabilities of harvesting robots. Currently, intelligent agricultural machinery equipped with sophisticated navigation and environmental sensing systems is becoming increasingly common in agricultural production. However, despite these advancements, the development of environmental perception technology for harvesting robots still faces a number of challenges that require resolution. The following is an in-depth discussion of four key aspects concerning the challenges and future directions of autonomous harvesting robots operational environmental perception technology.

4.1 Environmental sensing performance during picking robot

Accuracy and reaction time are the critical benchmarks for evaluating the environmental perception efficacy of harvesting robots. For accuracy, they leverage a range of sensors such as visual cameras, LiDAR, and radar to precisely detect and differentiate agricultural elements such as obstacles and crop lines, even under fluctuating light and weather conditions. In terms of reaction time, the system's performance hinges on both high-performance computing hardware and refined algorithms that can analyse incoming data streams promptly. This ensures that the robots can make rapid decisions and issue commands, completing tasks such as obstacle circumvention and route adjustment in milliseconds. Broadly speaking, environmental perception systems have made progress in terms of accurate recognition and swift response, yet they are not impervious to the practical challenges posed by fluctuating light conditions, uneven terrain, and obstacles. To address these issues and maintain a suitable safety standard for unmanned harvesting robots, it is imperative that the perception system's design incorporates a fail-safe threshold. This threshold should reflect the permissible period of dysfunction in perception before the robot's braking mechanism is triggered, steering the robot into a safe operational mode. Given the inherent constraints on the robots' line of sight, which hinders their ability to rapidly assimilate wide-ranging environmental cues, the incorporation of aerospace and aviation remote sensing technologies can be instrumental. These technologies are adept at pinpointing the static features of farmland. The aerial vantage point offered by drones, in particular, can significantly expedite the environmental modelling process for ground-based harvesting robots, thereby bolstering their capacity for real-time mapping and navigation in unfamiliar settings. Additionally, the proactive merging of harvesting robots with established agricultural methodologies can streamline the

(a) RGB data (b) Point clouds (c) Multisensor fusion

Fig. 6 Multisensor fusion (Colored figures are available in the online version at https://link.springer.com/journal/11633) operational milieu, thereby augmenting the precision of environmental perception for autonomous harvesting robots and mitigating inherent unpredictability. This convergence of technological innovation with agricultural know-how is set to forge a path toward a more productive and dependable model of agricultural automation.

4.2 Multisource sensor data fusion

On their own, sensors are constrained by factors such as the detection scope, signal attributes, and operational conditions, which often fall short in fulfilling the comprehensive environmental sensing needs of robotic harvesting systems. This has spurred the innovation of methodologies that integrate data from multiple sensors and synthesize their collective functionality. The integration of environmental perception data can be broadly divided into three methodologies: Gaussian filter-based estimators such as the Kalman filter and particle filters; probabilistic inference methods using Bayesian principles; and machine learning-driven artificial intelligence techniques. In autonomous vehicle technology, the prevalent strategy for sensor integration is to pair visual sensors with millimeter-wave radar, which offers the dual advantages of detailed distance measurements and high-resolution imaging while keeping costs in check. For the selection of sensors for robotic harvesting, it is crucial to consider a range of elements, including the detection range, sensitivity to light conditions, ability to function under poor atmospheric conditions, spatial resolution, and economic viability. The fusion process of data from various sensors is depicted in Fig. 4. Despite existing challenges, the technology is a promising trajectory for advancements. Enhancements in sensor fusion algorithms are anticipated to bolster the precision and dependability of environmental sensing. The advent of next-generation sensors promises enhanced resolution, extended detection capabilities, and robust resistance to interference, thus enriching the quality of data for fusion purposes. The impending standardization of environmental perception data in the realm of robotic harvesting is likely to foster greater system interoperability and data integration. Additionally, the pure visual perception approach from the autonomous driving sector offers significant lessons for harvesting robots via environmental sensing technology. It has been demonstrated that with accurate environmental mapping and sophisticated algorithms, a setup relying solely on multiple visual sensors can achieve superior navigation and obstacle avoidance. This underscores the significance of algorithmic refinement over the mere addition of sensor types.

4.3 Machine vision and mechanical fault tolerance

Robotic harvesters frequently function amidst the complexities of nature, where they encounter numerous external factors that can introduce substantial variability in the accuracy of their vision-based positioning systems. The patterns of these errors are often elusive, making them difficult to predict and correct. As a result, the integration of machine vision with mechanical fault tolerance has become a prominent area of study. Traditional vision systems usually overlook the interplay between mechanical and visual errors, focusing only on hardware-induced inaccuracies. In the realm of image processing and network training, the task is to segregate the target fruits from their surroundings, a process that requires meticulous error calculations and an understanding of how these errors can spread. Errors emerge not only from the image processing itself but also from the inherent noise that can skew computational results. The transmission of these errors through computational steps is a critical yet often understated aspect, highlighting the need for a coordinated approach between vision systems and mechanical components to address a spectrum of error types. There is a need for more in-depth research into the main contributing factors, including computational inaccuracies, stochastic errors, and their potential to propagate. Moreover, the presence of various interferences in the working environment can lead to significant random positioning errors for harvesting robots, which are difficult to rectify using conventional control algorithms. Current end effectors lack the capability to tolerate such errors, which are deemed systemic "fault errors". To counter this, innovative end effector designs, such as those proposed by researchers, including Xu et al.[102], have been introduced to counteract these random errors. These new end effectors, when combined with RGB-D cameras on a mobile platform, are fitted with internal sensors that can detect and adjust for positioning discrepancies within the vision system. They also demonstrate resilience against the errors imparted by the vision module. In the domain of harvesting robot research, it is imperative to investigate fault-tolerant theories to ensure the accurate positioning of target fruits using machine vision and to bolster the overall robustness of these systems against errors.

4.4 Outlook

Target recognition and 3D positioning of picking robots have always been popular topics in the field of agricultural machinery research. Compared with traditional recognition and positioning methods, deep learning-based convolutional neural networks are more worthy of attention. From the current development status of the research on target recognition and positioning methods for vision-based picking robots, the following aspects of work are still worthy of attention:

1) Algorithm optimization. Algorithm optimization in the vision system of a picking robot involves continuous implementation of existing target recognition and positioning algorithms to improve accuracy and robustness in

a changing agricultural environment. This includes using advanced deep learning techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to automatically extract crop features and enhance the model′s resilience to light changes, occlusion and seasonal differences while reducing computational resource consumption through algorithm-based reduction and model compression techniques. The robots can perform tasks efficiently and in real time under resource-constrained conditions.

2) Sensor development. Sensor development in vision systems for picking robots refers to the development and refinement of multiple sensing technologies for the precise identification and positioning of ripe fruits, including but not limited to high-resolution cameras, 3D depth sensors, multispectral and near-infrared sensors, and possibly haptic feedback devices. The integration and optimization of these sensors are designed to improve the robot′s adaptability to different crop characteristics, ambient light conditions and complex backgrounds, resulting in more accurate and reliable target detection, even when the fruit is partially obscured or when the light conditions are not ideal. In addition, the research and development work includes improving the durability of the sensor, reducing costs, and enhancing its stability under all weather conditions.

3) Mechanical fault tolerance. The application of machine vision in picking robots involves the use of advanced image processing technology and pattern recognition algorithms to enable robots to accurately identify and locate fruits, while mechanical fault tolerance refers to the integration of intelligent algorithms to improve adaptability to unexpected situations when designing a robot′s mechanical structure and control system. Even in the face of complex situations such as fruit overload, occlusion or environmental changes, a robot can operate stably and adjust itself, reducing errors and potential damage, thereby improving overall picking efficiency and system reliability.

5 Conclusions

The perception of the environment is essential for the functionality of harvesting robots. This discussion encompasses a range of sensor technologies and image processing strategies vital for the detection and positioning of horticultural produce, evaluating its precision, velocity, and resilience. The efficacy of these visual systems in identifying and locating fruits and vegetables depends on the specific crop type and is subject to a variety of influences, including light conditions and obstructions. The discourse synthesizes the current state of research on the environmental perception technologies utilized by developing robots, probing the challenges inherent in the field and the prospective trajectories of its evolution. Given the intricate and ever-shifting landscape of agricultural settings, which are inherently unstructured, these systems require heightened reliability and swift response times. There is a pressing need for significant research advancements in sensor capabilities for environmental purposes, the amalgamation of data from diverse sensors, and the enhancement of machine vision alongside mechanical fault tolerance. Progress in these technical and methodological domains is crucial for the ongoing innovation and improvement of environmental perception technologies in harvesting robots, thereby reinforcing the drive towards smart agriculture.

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Declarations of conflict of interest

Xiao Xu conceptualized and designed the algorithm, implemented the initial codebase, and prepared the original manuscript draft.

Yaonan Wang contributed to the development and fine-tuning of the algorithm, performed substantial debugging and code optimization, and assisted with manuscript writing and revisions.

Yiming Jiang designed and executed the performance tests, analysed the computational results, and contributed to the interpretation of these results in the manuscript.

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References

[1] Y. G. Liu, B. Y. Li, Q. Fu, X. M. Zhang, F. Y. Ma, Y. Hu. Miracle fruit leaf extract: Antioxidant activity evaluation, constituent identification, and medical applications. *Analytical Letters*, vol. 54, no. 13, pp. 2211–2226, 2021. DOI: 10.1080/00032719.2020.1854277.

[2] Y. Park, C. Kim, H. I. Son. Fast and stable pedicel detection for robust visual servoing to harvest shaking fruits. *Computers and Electronics in Agriculture*, vol. 220, Article number 108863, 2024. DOI: 10.1016/j.compag.2024.108863.

[3] M. Liu, F. Wang, X. P. Wang, Y. N. Wang, A. K. Roy-Chowdhury. A two-stage noise-tolerant paradigm for label corrupted person re-identification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 7, pp. 4944–4956, 2024. DOI: 10.1109/TPAMI.2024.3361491.

[4] S. H. Lee, Y. J. Lee, D. H. Kim. Extension of inverse kinematic solution for a robot to cope with joint angle constraints. *International Journal of Control, Automation and Systems*, vol. 21, no. 6, pp. 1899–1909, 2023. DOI: 10.1007/s12555-021-1052-6.

[5] B. Q. Sun, R. X. Jing, Z. S. Wang, L. Tian, F. F. Mao, Y. Liu. Diversity and community structure of endophytic *Bacillus* with antagonistic and antioxidant activity in the fruits of Xisha Wild Noni (*Morinda citrifolia* L.). *Microbial Pathogenesis*, vol. 158, Article number 105065, 2021. DOI: 10.1016/j.micpath.2021.105065.

[6] Z. J. Mu, X. L. Du, X. Y. Wang. Identifying internal control genes by quantitative real-time PCR for accurate normalisation of gene expression in *Gardenia Jasminoides*. *New Zealand Journal of Crop and Horticultural Science*, vol. 49, no. 4, pp. 305–324, 2021. DOI: 10.1080/01140671.2021.1879176.

[7] J. S. Liu, G. Z. Zhao, S. X. Liu, Y. Liu, H. W. Yang, J. W. Sun, Y. F. Yan, G. Q. Fan, J. X. Wang, H. J. Zhang. New progress in intelligent picking: Online detection of apple maturity and fruit diameter based on machine vision. *Agronomy*, vol. 14, no. 4, Article number 721, 2024. DOI: 10.3390/agronomy14040721.

[8] H. J. Huang, D. Wang, A. Tangri, R. Walters, R. Platt. Leveraging symmetries in pick and place. *The International Journal of Robotics Research*, vol. 43, no. 4, pp. 550–571, 2024. DOI: 10.1177/02783649231225775.

[9] M. Y. Chen, Z. X. Chen, L. F. Luo, Y. C. Tang, J. B. Cheng, H. L. Wei, J. H. Wang. Dynamic visual servo control methods for continuous operation of a fruit harvesting robot working throughout an orchard. *Computers and Electronics in Agriculture*, vol. 219, Article number 108774, 2024. DOI: 10.1016/j.compag.2024.108774.

[10] J. Zhang, N. B. Kang, Q. J. Qu, L. H. Zhou, H. B. Zhang. Automatic fruit picking technology: A comprehensive review of research advances. *Artificial Intelligence Review*, vol. 57, no. 3, Article number 54, 2024. DOI: 10.1007/s10462-023-10674-2.

[11] Q. W. Li, W. K. Jia, M. L. Sun, S. J. Hou, Y. J. Zheng. A novel green apple segmentation algorithm based on ensemble U-Net under complex orchard environment. *Computers and Electronics in Agriculture*, vol. 180, Article number 105900, 2021. DOI: 10.1016/j.compag.2020.105900.

[12] D. H. Wu, Q. Wu, X. Q. Yin, B. Jiang, H. Wang, D. J. He, H. B. Song. Lameness detection of dairy cows based on the YOLOv3 deep learning algorithm and a relative step size characteristic vector. *Biosystems Engineering*, vol. 189, pp. 150–163, 2020. DOI: 10.1016/j.biosystemseng.2019.11.017.

[13] J. T. Xiong, Z. Liu, S. M. Chen, B. L. Liu, Z. H. Zheng, Z. Zhong, Z. G. Yang, H. X. Peng. Visual detection of green mangoes by an unmanned aerial vehicle in orchards based on a deep learning method. *Biosystems Engineering*, vol. 194, pp. 261–272, 2020. DOI: 10.1016/j.biosystemseng.2020.04.006.

[14] K. M. Alaaudeen, S. Selvarajan, H. Manoharan, R. H. Jhaveri. Intelligent robotics harvesting system process for fruits grasping prediction. *Scientific Reports*, vol. 14, no. 1, Article number 2820, 2024. DOI: 10.1038/s41598-024-52743-8.

[15] W. G. Ding, G. Taylor. Automatic moth detection from trap images for pest management. *Computers and Electronics in Agriculture*, vol. 123, pp. 17–28, 2016. DOI: 10.1016/j.compag.2016.02.003.

[16] Q. Lü, J. R. Cai, B. Liu, L. Deng, Y. J. Zhang. Identification of fruit and branch in natural scenes for citrus harvesting robot using machine vision and support vector machine. *International Journal of Agricultural and Biological Engineering*, vol. 7, no. 2, pp. 115–121, 2014. DOI: 10.3965/j.ijabe.20140702.014.

[17] M. Rahnemoonfar, C. Sheppard. Deep count: Fruit counting based on deep simulated learning. *Sensors*, vol. 17, no. 4, Article number 905, 2017. DOI: 10.3390/s17040905.

[18] M. Stein, S. Bargoti, J. Underwood. Image based mango fruit detection, localisation and yield estimation using multiple view geometry. *Sensors*, vol. 16, no. 11, Article number 1915, 2016. DOI: 10.3390/s16111915.

[19] J. Wang, Y. N. Li, H. L. Feng, L. J. Ren, X. C. Du, J. Wu. Common pests image recognition based on deep convolutional neural network. *Computers and Electronics in Agriculture*, vol. 179, Article number 105834, 2020. DOI: 10.1016/j.compag.2020.105834.

[20] X. X. Sun, S. M. Mu, Y. Y. Xu, Z. H. Cao, T. T. Su. Image recognition of tea leaf diseases based on convolutional neural network. In *Proceedings of International Conference on Security, Pattern Analysis, and Cybernetics*, Jinan, China, pp. 304–309, 2018. DOI: 10.1109/SPAC46244.2018.8965555.

[21] G. F. Lin, W. Shen. Research on convolutional neural network based on improved Relu piecewise activation function. *Procedia Computer Science*, vol. 131, pp. 977–984, 2018. DOI: 10.1016/j.procs.2018.04.239.

[22] S. J. Jia, P. Wang, P. Y. Jia, S. P. Hu. Research on data augmentation for image classification based on convolution neural networks. In *Proceedings of Chinese Automation Congress*, Jinan, China, pp. 4165–4170, 2017. DOI: 10.1109/CAC.2017.8243510.

[23] V. Khomenko, O. Shyshkov, O. Radyvonenko, K. Bokhan. Accelerating recurrent neural network training using sequence bucketing and multi-GPU data parallelization. In *Proceedings of the IEEE First International*

Conference on Data Stream Mining & Processing, Lviv, Ukraine, pp. 100–103, 2016. DOI: 10.1109/DSMP.2016.7583516.

[24] P. A. Dias, A. Tabb, H. Medeiros. Apple flower detection using deep convolutional networks. *Computers in Industry*, vol. 99, pp. 17–28, 2018. DOI: 10.1016/j.compind.2018.03.010.

[25] Z. S. Han, C. P. Wang, Q. Fu. Arbitrary-oriented target detection in large scene sar images. *Defence Technology*, vol. 16, no. 4, pp. 933–946, 2020. DOI: 10.1016/j.dt.2019.11.014.

[26] X. Zhang, Q. M. Gao, D. Pan, P. C. Cao, D. H. Huang. Research on spatial positioning system of fruits to be picked in field based on binocular vision and SSD model. *Journal of Physics: Conference Series*, vol. 1748, Article number 042011, 2021. DOI: 10.1088/1742-6596/1748/4/042011.

[27] H. W. Kang, C. Chen. Fruit detection, segmentation and 3D visualisation of environments in apple orchards. *Computers and Electronics in Agriculture*, vol. 171, Article number 105302, 2020. DOI: 10.1016/j.compag.2020.105302.

[28] H. W. Kang, H. Y. Zhou, C. Chen. Visual perception and modeling for autonomous apple harvesting. *IEEE Access*, vol. 8, pp. 62151–62163, 2020. DOI: 10.1109/ACCESS.2020.2984556.

[29] J. Park, C. Han, M. B. G. Jun, H. Yun. Autonomous robotic bin picking platform generated from human demonstration and YOLOv5. *Journal of Manufacturing Science and Engineering*, vol. 145, no. 12, Article number 121006, 2023. DOI: 10.1115/1.4063107.

[30] H. Kang, H. Y. Zhou, X. Wang, C. Chen. Real-time fruit recognition and grasping estimation for robotic apple harvesting. *Sensors*, vol. 20, no. 9, Article number 5670, 2020. DOI: 10.3390/s20195670.

[31] M. Y. Chen, Y. C. Tang, X. J. Zou, K. Y. Huang, Z. F. Huang, H. Zhou, C. L. Wang, G. P. Lian. Three-dimensional perception of orchard banana central stock enhanced by adaptive multi-vision technology. *Computers and Electronics in Agriculture*, vol. 174, Article number 105508, 2020. DOI: 10.1016/j.compag.2020.105508.

[32] X. Ling, Y. S. Zhao, L. Gong, C. L. Liu, T. Wang. Dual-arm cooperation and implementing for robotic harvesting tomato using binocular vision. *Robotics and Autonomous Systems*, vol. 114, pp. 134–143, 2019. DOI: 10.1016/j.robot.2019.01.019.

[33] D. Beale, P. Iravani, P. Hall. Statistical visual-dynamic model for hand-eye coordination. In *Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems*, Taipei, China, pp. 3931–3936, 2010. DOI: 10.1109/IROS.2010.5648832.

[34] F. Sistler. Robotics and intelligent machines in agriculture. *IEEE Journal on Robotics and Automation*, vol. 3, no. 1, pp. 3–6, 1987. DOI: 10.1109/JRA.1987.1087074.

[35] T. Grift, Q. Zhang, N. Kondo, K. C. Ting. A review of automation and robotics for the bioindustry. *Journal of Biomechatronics Engineering*, vol. 1, no. 1, pp. 37–54, 2008.

[36] S. Arivazhagan, R. N. Shebiah, S. S. Nidhyanandhan, L. Ganesan. Fruit recognition using color and texture features. *Journal of Emerging Trends in Computing and Information Sciences*, vol. 1, no. 2, pp. 90–94, 2010.

[37] A. R. Jiménez, A. K. Jain, R. Ceres, J. L. Pons. Automatic fruit recognition: A survey and new results using range/attenuation images. *Pattern Recognition*, vol. 32, no. 10, pp. 1719–1736, 1999. DOI: 10.1016/S0031-3203(98)00170-8.

[38] L. Y. Yu, J. T. Xiong, X. Q. Fang, Z. G. Yang, Y. Q. Chen, X. Y. Lin, S. F. Chen. A litchi fruit recognition method in a natural environment using RGB-D images. *Biosystems Engineering*, vol. 204, pp. 50–63, 2021. DOI: 10.1016/j.biosystemseng.2021.01.015.

[39] T. Ilyas, A. Khan, M. Umraiz, Y. Jeong, H. Kim. Multi-scale context aggregation for strawberry fruit recognition and disease phenotyping. *IEEE Access*, vol. 9, pp. 124491–124504, 2021. DOI: 10.1109/ACCESS.2021.3110978.

[40] S. H. Tan, C. K. Lam, K. Kamarudin, A. H. Ismail, N. A. Rahim, M. S. M. Azmi, W. M. N. W. Yahya, G. K. Sneah, M. L. Seng, T. P. Hai, O. T. Lye. Vision-based edge detection system for fruit recognition. *Journal of Physics: Conference Series*, vol. 2107, Article number 012066, 2021. DOI: 10.1088/1742-6596/2107/1/012066.

[41] J. D. Lv, X. J. Lv, Z. H. Ma. A fruit recognition method of green grape images in the orchard. *New Zealand Journal of Crop and Horticultural Science*, vol. 50, no. 1, pp. 1–16, 2022. DOI: 10.1080/01140671.2021.1885451.

[42] H. H. C. Nguyen, A. T. Luong, T. H. Trinh, P. H. Ho, P. Meesad, T. Thuy Nguyen. Intelligent fruit recognition system using deep learning. *Recent Advances in Information and Communication Technology 2021*, P. Meesad, D. S. Sodsee, W. Jitsakul, S. Tangwannawit, Eds., Cham, Germany: Springer, pp. 13–22, 2021. DOI: 10.1007/978-3-030-79757-7_2.

[43] D. Q. Xu, L. T. Chen, X. W. Mou, Q. Wu, G. Q. Sun. 3D reconstruction of camellia oleifera fruit recognition and fruit branch based on Kinect camera. In *Proceedings of the 2nd International Conference on Artificial Intelligence and Information Systems*, Chongqing, China, Article number 118, 2021. DOI: 10.1145/3469213.3470321.

[44] H. Naidu, S. Rajkumar, K. C. Santosh, P. V. S. S. R. Chandra Mouli. A fast and efficient convolutional neural network for fruit recognition and classification. *Recent Trends in Image Processing and Pattern Recognition*, K. C. Santosh, B. Gawali, Eds., Singapore: Springer, pp. 148–157, 2021. DOI: 10.1007/978-981-16-0507-9_13.

[45] S. Harini, P. Deshpande, J. Dutta, B. Rai. A deep learning-based fruit quality assessment system. In *Proceedings of the 1st International Conference on Water Energy Food and Sustainability*, pp. 187–192, 2021. DOI: 10.1007/978-3-030-75315-3_22.

[46] R. Wan Nurazwin Syazwani, A. H. Muhammad, M. A. Megat Syahirul Amin, K. A. Nur Dalila. Automated image identification, detection and fruit counting of top-view pineapple crown using machine learning. *Alexandria Engineering Journal*, vol. 61, no. 2, pp. 1265–1276, 2022. DOI: 10.1016/j.aej.2021.06.053.

[47] X. B. Zhang, Y. H. Zhu, Y. L. Su, B. L. Xie, Q. Gu, K. F. Zheng. Quantitative extraction and analysis of pear fruit spot phenotypes based on image recognition. *Computers and Electronics in Agriculture*, vol. 190, Article number 106474, 2021. DOI: 10.1016/j.compag.2021.106474.

[48] K. M. Alresheedi, S. Aladhadh, R. U. Khan, A. M. Qamar. Dates fruit recognition: From classical fusion to deep learning. *Computer Systems Science and Engineering*, vol. 40, no. 1, pp. 151–166, 2022. DOI: 10.32604/csse.2022.017931.

[49] M. Yang, P. Kumar, J. Bhola, M. Shabaz. Development of image recognition software based on artificial intelligence algorithm for the efficient sorting of apple fruit. *International Journal of System Assurance Engineering and Management*, vol. 13, no. 1S, pp. 322–330, 2022. DOI: 10.1007/s13198-021-01415-1.

[50] K. R. Jerripothula, S. K. Shukla, S. Jain, S. Singh. Fruit maturity recognition from agricultural, market and automation perspectives. In *Proceedings of the 47th Annual Conference of the IEEE Industrial Electronics Society*, Toronto, Canada, 2021. DOI: 10.1109/IECON48115.2021.9589215.

[51] M. A. Fadhel, O. Al-Shamma. Implementing a hardware accelerator to enhance the recognition performance of the fruit mature. *Recent Trends in Signal and Image Processing: ISSIP 2020*, S. Bhattacharyya, L. Mršić, M. Brkljačić, J. V. Kureethara, M. Koeppen, Eds., Singapore: Springer, pp. 41–52, 2021. DOI: 10.1007/978-981-33-6966-5_5.

[52] F. A. Shah, M. A. Khan, M. Sharif, U. Tariq, A. Khan, S. Kadry, O. Thinnukool. A cascaded design of best features selection for fruit diseases recognition. *Computers, Materials & Continua*, vol. 70, no. 1, pp. 1491–1507, 2022. DOI: 10.32604/cmc.2022.019490.

[53] X. L. Zhou, P. B. Wang, G. L. Dai, J. W. Yan, Z. Yang. Tomato fruit maturity detection method based on YOLOV4 and statistical color model. In *Proceedings of the IEEE 11th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems*, Jiaxing, China, pp. 904–908, 2021. DOI: 10.1109/CYBER53097.2021.9588129.

[54] A. K. Kleih, K. Sparke. Visual marketing: The importance and consumer recognition of fruit brands in supermarket fruit displays. *Food Quality and Preference*, vol. 93, Article number 104263, 2021. DOI: 10.1016/j.foodqual.2021.104263.

[55] J. Pardede, B. Sitohang, S. Akbar, M. L. Khodra. Implementation of transfer learning using VGG16 on fruit ripeness detection. *International Journal of Intelligent Systems and Applications* (*IJISA*), vol. 13, no. 2, pp. 52–61, 2021. DOI: 10.5815/ijisa.2021.02.04.

[56] J. H. Yu, W. J. Miao, G. B. Zhang, K. Li, Y. G. Shi, L. Liu. Target positioning and sorting strategy of fruit sorting robot based on image processing. *Traitement du Signal*, vol. 38, no. 3, pp. 797–805, 2021. DOI: 10.18280/ts.380326.

[57] A. Arefi, A. M. Motlagh, K. Mollazade, R. F. Teimourlou. Recognition and localization of ripen tomato based on machine vision. *Australian Journal of Crop Science*, vol. 5, pp. 1144–1149, 2011.

[58] R. J. Mhaske, S. B. Dabhade, S. Mache, K. S. Rana, P. Deshmukh. Periodical fruit quality identification – A broad view. *ICCCE 2020*, A. Kumar, S. Mozar, Eds., Singapore: Springer, pp. 679–688, 2021. DOI: 10.1007/978-981-15-7961-5_65.

[59] G. X. Liu, S. Y. Mao, J. H. Kim. A mature-tomato detection algorithm using machine learning and color analysis. *Sensors*, vol. 19, no. 9, 2023. DOI: 10.3390/s19092023.

[60] W. Ni, R. H. Li. Research on recognition technology of fruit based on simulated annealing algorithm and neural network. In *Proceedings of IEEE Asia-Pacific Conference on Image Processing, Electronics and Computers*, Dalian, China, pp. 877–880, 2021. DOI: 10.1109/IPEC51340.2021.9421261.

[61] Y. Y. Wei, L. H. Cai, H. F. Fang, H. Y. Chen. Fruit recognition and classification based on tactile information of flexible hand. *Sensors and Actuators A: Physical*, vol. 370, 2024. DOI: 10.1016/j.sna.2024.115224.

[62] X. Xiao, Y. Wang, B. Zhou, Y. Jiang. Flexible hand claw picking method for citrus-picking robot based on target fruit recognition. *Agriculture*, vol. 14, no.8, Article number 1227, 2024. https://doi.org/10.3390/agriculture14081227

[63] M. T. Habib, M. A. I. Arif, S. B. Shorif, M. S. Uddin, F. Ahmed. Machine vision-based fruit and vegetable disease recognition: A review. *Computer Vision and Machine Learning in Agriculture*, M. S. Uddin, J. C. Bansal, Eds., Singapore: Springer, pp. 143–157, 2021. DOI: 10.1007/978-981-33-6424-0_10.

[64] S. K. Behera, P. K. Sethy, S. K. Sahoo, S. Panigrahi, S. C. Rajpoot. On-tree fruit monitoring system using IoT and image analysis. *Concurrent Engineering*, vol. 29, no. 1, pp. 6–15, 2021. DOI: 10.1177/1063293X20988395.

[65] A. Oualid Djekoune, K. Messaoudi, K. Amara. Incremental circle hough transform: An improved method for circle detection. *Optik*, vol. 133, pp. 17–31, 2017.

[66] Q. Xie, M. T. Luong, E. Hovy and Q. V. Le. Self-training with noisy student improves imagenet classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, Seattle, USA, 2020.

[67] Y. Tene-Cohen, T. Kelman, O. Lev, A. Makmal. A variational qubit-efficient MaxCut heuristic algorithm, [Online], Available: https://arxiv.org/abs/2308.10383.

[68] J. L. Joseph, V. A. Kumar, S. P. Mathew. Fruit classification using deep learning. *Innovations in Electrical and Electronic Engineering*, S. Mekhilef, M. Favorskaya, R. K. Pandey, R. N. Shaw, Eds., Singapore: Springer, pp. 807–817, 2021. DOI: 10.1007/978-981-16-0749-3_62.

[69] Z. Zhong, J. T. Xiong, Z. H. Zheng, B. L. Liu, S. S. Liao, Z. W. Huo, Z. G. Yang. A method for litchi picking points calculation in natural environment based on main fruit bearing branch detection. *Computers and Electronics in Agriculture*, vol. 189, Article number 106398, 2021. DOI: 10.1016/j.compag.2021.106398.

[70] F. Kurtulmus, W. S. Lee, A. Vardar. Green citrus detection using 'eigenfruit', color and circular Gabor texture features under natural outdoor conditions. *Computers and Electronics in Agriculture* , vol. 78, no. 2, pp. 140–149, 2011.

[71] L. J. Qi, J. R. Zhou, J. J. Wan, Z. P. Yang, H. Zhang, Z. Z. Cheng. Canopy recognition of cherry fruit tree based on SegNet network model. In *Proceedings of SPIE 11915, International Conference on Optics and Image Processing*, Guilin, China, pp. 92–104, 2021. DOI: 10.1117/12.2605881.

[72] M. Nuruzzaman, M. S. Hossain, M. M. Rahman, A. S. H. C. Shoumik, M. A. A. Khan, M. T. Habib. Machine vis-

ion based potato species recognition. In *Proceedings of the 5th International Conference on Intelligent Computing and Control Systems*, Madurai, India, 2021. DOI: 10.1109/ICICCS51141.2021.9432195.

[73] A. W. R. Payne, R. C. Glen. Molecular recognition using a binary genetic search algorithm. *Journal of Molecular Graphics*, vol. 11, no. 2, pp. 74–91, 1993.

[74] G. Khekare, K. Wankhade, U. Dhanre, B. Vidhale. Internet of things based best fruit segregation and taxonomy system for smart agriculture. *IoT and Cloud Computing for Societal Good*, J. K. Verma, D. Saxena, V. González-Prida, Eds., Cham, Germany: Springer, pp. 37–57, 2022. DOI: 10.1007/978-3-030-73885-3_4.

[75] B. Y. Wang, Y. X. Xie, X. H. Duan. An IoT based fruit and vegetable sales system: A whole system including IoT based integrated intelligent scale and online shop. In *Proceedings of the 5th International Conference on Cloud and Big Data Computing*, Liverpool, UK, pp. 109–115, 2021. DOI: 10.1145/3481646.3481663.

[76] M. H. Abd Alkreem, A. A. Karim. Feature selection using artificial bee colony for fruit classification. *Journal of Physics: Conference Series*, vol. 1818, Article number 012062, 2021. DOI: 10.1088/1742-6596/1818/1/012062.

[77] A. Chougule, A. Pawar, R. Kamble, J. Mujawar, A. Bhide. Recognizing fresh and rotten fruits using deep learning techniques. *Data Engineering and Intelligent Computing*, V. Bhateja, S. C. Satapathy, C. M. Travieso-González, V. N. Manjunath Aradhya, Eds., Singapore: Springer, pp. 205–212, 2021. DOI: 10.1007/978-981-16-0171-2_20.

[78] Y. L. Chen, H. L. Sun, G. F. Zhou, B. Peng. Fruit classification model based on residual filtering network for smart community robot. *Wireless Communications and Mobile Computing*, vol. 2021, Article number 5541665, 2021. DOI: 10.1155/2021/5541665.

[79] M. T. Habib, A. Khatun, M. A. Aziz, M. S. Uddin, F. Ahmed. An in-depth analysis of different segmentation techniques in automated local fruit disease recognition. *Computer Vision and Machine Learning in Agriculture*, M. S. Uddin, J. C. Bansal, Eds., Singapore: Springer, pp. 127–142, 2021. DOI: 10.1007/978-981-33-6424-0_9.

[80] Y. H. Zhan, M. Y. Chen, Y. F. Chen, Z. Q. Luo, Y. Luo, K. D. Li. An automatic recognition method of fruits and vegetables based on depthwise separable convolution neural network. *Journal of Physics: Conference Series*, vol. 1871, Article number 012075, 2021. DOI: 10.1088/1742-6596/1871/1/012075.

[81] P. Maheswari, P. Raja, O. E. Apolo-Apolo, M. Pérez-Ruiz. Intelligent fruit yield estimation for orchards using deep learning based semantic segmentation techniques – A review. *Frontiers in Plant Science*, vol. 12, Article number 684328, 2021. DOI: 10.3389/fpls.2021.684328.

[82] A. S. Rao, M. P. Kumar, C. H. Jagadeesh, G. Yoganand. An advanced android based trained fruit and vegetable identification nutrition. *Journal of Engineering Sciences*, vol. 12, no. 7, pp. 293–299, 2021.

[83] M. Yamamoto. Clustering of functional data in a low-dimensional subspace. *Advances in Data Analysis and Classification*, vol. 6, pp. 219–247, 2012.

[84] F. Xiao, H. Wang, Y. Xu, R. Q. Zhang. Fruit detection and recognition based on deep learning for automatic harvesting: an overview and review. *Agronomy*, vol. 13, no. 6, 2023.

[85] R. Luo, Y. Tian, W. Jia, Z. Zhang, J. Lian, Y. Zheng. Detection and segmentation of overlapped fruits based on optimized mask R-CNN application in apple harvesting robot. *Computers and Electronics in Agriculture*, vol. 172, Article number 105380, 2020. DOI: 10.1016/j.compag.2020.105380.

[86] X. N. Li, H. Wu, X. H. Yang. Multi-view recognition of fruit packing boxes based on features clustering angle. *High Technology Letters*, vol. 27, no. 2, pp. 200–209, 2021. DOI: 10.3772/j.issn.1006-6748.2021.02.011.

[87] L. Wang, X. Wu, Z. Qi, J. Yang, X. Xia. Apple detection method based on light-YOLOv3 convolutional neural network. *Transactions of the Chinese Society for Agricultural Machinery*. DOI: 10.6041/j.issn.1000-1298.2020.08.002.

[88] P. Moallem, A. Serajoddin, H. Pourghassem. Computer vision-based apple grading for golden delicious apples based on surface features. *Information Processing in Agriculture*, vol. 4, no. 1, pp. 33–40, 2017. DOI: 10.1016/j.inpa.2016.10.003.

[89] J. J. Bird, C. M. Barnes, L. J. Manso, A. Ekárt, D. R. Faria. Fruit quality and defect image classification with conditional GAN data augmentation. *Scientia Horticulturae*, vol. 293, Article number 110684, 2022. DOI: 10.1016/j.scienta.2021.110684.

[90] R. C. Harrell, D. C. Slaughter, P. D. Adsit. A fruit-tracking system for robotic harvesting, *Machine Vision and Applications*, vol. 2, pp. 69–80, 1989.

[91] W. Min, Z. Wang, J. Yang, C. Liu, S. Jiang. Vision-based fruit recognition via multi-scale attention CNN. *Computers and Electronics in Agriculture*, vol. 210, Article number 107911, 2023.

[92] A. Gao, Z. Tian, W. Ma, Y. Song, L. Ren, Y. Feng, J. Qian, L. Xu. Fruits hidden by green: an improved YOLOV8n for detection of young citrus in lush citrus trees. *Frontiers in Plant Science*, vol. 15, 2024. DOI: 10.3389/fpls.2024.1375118.

[93] H. S. Gill, G. Murugesan, B. S. Khehra, G. S. Sajja, G. Gupta, A. Bhatt. Fruit recognition from images using deep learning applications. *Multimedia Tools and Applications*, vol. 81, pp. 33269–33290, 2022.

[94] H. Peng, C. Xue, Y. Shao, K. Chen, J. Xiong, Z. Xie. Semantic segmentation of litchi branches using DeepLabV3 + Model. *IEEE Access*, vol. 8, pp. 164546–164555, 2020. DOI: 10.1109/ACCESS.2020.3021739.

[95] L. T. Duong, P. T. Nguyen, C. Di Sipio, D. Di Ruscio. Automated fruit recognition using EfficientNet and MixNet. *Computers and Electronics in Agriculture*, vol. 171, Article number 105326, 2020. DOI: 10.1016/j.compag.2020.105326.

[96] S. I. Saedi, H. Khosravi. A deep neural network approach towards real-time on-branch fruit recognition for precision horticulture. *Expert Systems with Applications*, vol. 159, Article number 113594, 2020. DOI: 10.1016/j.eswa.2020.113594.

[97] H. Peng, Y. Shao, K. Chen, Y. Deng, C. Xue. Research on multi-class fruits recognition based on machine vision and SVM. *IFAC-PapersOnLine*, vol. 51, no. 17, pp. 817–821, 2018.

[98] F. Xiao, H. Wang, Y. Li, Y. Cao, X. Lv, G. Xu. Object detection and recognition techniques based on digital image processing and traditional machine learning for fruit and vegetable harvesting robots: An overview and review. *Agronomy*, vol. 13, no. 3, 2023. DOI: 10.3390/agronomy13030639.

[99] G. Wu, B. Li, Q. B. Zhu, M. Huang, Y. Guo. Using color and 3D geometry features to segment fruit point cloud and improve fruit recognition accuracy. *Computers and Electronics in Agriculture*, vol. 174, Article number 105475, 2020. DOI: 10.1016/j.compag.2020.105475.

[100] T. Sun, W. Zhang, Z. H. Miao, Z. Zhang, N. Li. Object localization methodology in occluded agricultural environments through deep learning and active sensing. *Computers and Electronics in Agriculture*, vol. 212, Article number 108141, 2023. DOI: 10.1016/j.compag.2023.108141.

[101] W. K. Jia, S. H. Mou, J. Wang, X. Y. Liu, Y. J. Zheng, J. Lian, D. A. Zhao. Fruit recognition based on pulse coupled neural network and genetic Elman algorithm application in apple harvesting robot. *International Journal of Advanced Robotic Systems*, vol. 17, no. 1, Article number 1729881419897473, 2020. DOI: 10.1177/1729881419897473.

[102] X. Xu, Y. M. Jiang, Y. N. Wang. A method of robot picking citrus based on 3D detection. *IEEE Instrumentation & Measurement Magazine*, vol. 27, no. 3, pp. 50–58, 2024.

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